1. Plotting and Fitting of Binomial Distribution and Graphical Representation of Probabilities

The binomial distribution is one of the most fundamental discrete probability distributions in statistics. It is used to determine the likelihood of a specific number of successes occurring across a fixed number of independent trials, where every trial can yield one of only two possible outcomes — a success or a failure. This distribution operates under the assumption that each trial is statistically independent and that the probability of achieving a success remains unchanged throughout all trials.

The binomial distribution is characterized by two key parameters:

- *n: represents the total number of trials conducted*
- *p: denotes the probability of achieving a success in any individual trial*

Binomial Distribution Formula

```
P(X = k) = C(n, k) * p^k * (1 - p)^(n-k)
```

Mean: $\mu = E(X) = n * p$

Variance: $\sigma^2 = V(X) = n * p * (1 - p)$

Where:

- n = total number of independent trials
- k = number of successes observed
- p = probability of success on any given trial
- C(n, k) = n! / [k! * (n - k)!] (the binomial coefficient)

How to Implement in Excel

Excel Formula: =BINOM.DIST(k, n, p, FALSE)

The following tables display the binomial distribution probabilities for n = 10 trials under three different values of the success probability p.

Case 1: n = 10, p = 0.8 (Right-skewed; bulk of probability concentrated at higher values)

k (Successes)	Binomial Probability
0	0.00000102
1	0.00004096

k (Successes)	Binomial Probability
2	0.00073728
3	0.00786432
4	0.05505024
5	0.20132659
6	0.30198989
7	0.26843546
8	0.10737418
9	0.02684354
10	0.00268435

When p = 0.8, the distribution is skewed to the left. The bulk of probability mass is concentrated in the higher success counts, with the distribution tailing off toward the lower end.

Case 2: n = 10, p = 0.5 (Symmetric distribution)

k (Successes)	Binomial Probability
0	0.00097656
1	0.00976563
2	0.04394531
3	0.11718750
4	0.20507813
5	0.24609375
6	0.20507813
7	0.11718750
8	0.04394531
9	0.00976563
10	0.00097656

When p = 0.5, the distribution exhibits perfect symmetry around the central value, n/2 = 5.

Case 3: n = 10, p = 0.2 (Left-skewed; concentrated at lower success counts)

k (Successes)	Binomial Probability
0	0.10737418
1	0.26843546
2	0.30198989
3	0.20132659
4	0.08808038
5	0.02642412
6	0.00880804
7	0.00073728
8	0.00040960
9	0.00004096
10	0.00000410

When p = 0.2, the distribution is right-skewed, with the highest probabilities occurring at small values of k.

2. Plotting and Fitting of Multinomial Distribution and Graphical Representation of Probabilities

The multinomial distribution is a multivariate extension of the binomial distribution. It applies when an experiment can result in exactly one of k possible distinct outcomes per trial, with each outcome i having a fixed probability p_i (where the sum of all probabilities equals 1), and there are n statistically independent trials. The random variables X_1, X_2, ..., X_k each record how many times outcome i was observed across the n trials.

When the joint vector X = (X_1, X_2, ..., X_k) follows a multinomial distribution with parameters n and p = (p_1, p_2, ..., p_k), each individual X_i follows a marginal binomial distribution with n trials and success probability p_i.

Multinomial Distribution Formula

```
P(x1, x2, ..., xk) = [n! / (x1! * x2! * ... * xk!)] * p1^x1 *
                     p2^x2 * ... * pk^xk
```

```
Cov(Xi, Xj) = -n * pi * pj   (for i ≠ j)
```

How to Implement in Excel

Multinomial coefficient: =MULTINOMIAL(X1, X2, X3)

Probability: =MULTINOMIAL(X1,X2,X3) * PRODUCT(p1^X1, p2^X2, p3^X3)

The table below presents three different scenarios involving red, green, and blue balls drawn from a container:

Parameter	Case 1	Case 2	Case 3
Red (X1)	5	4	3
Green (X2)	1	2	1
Blue (X3)	4	4	6
P_Red (P1)	0.375	0.375	0.375
P_Green (P2)	0.125	0.125	0.125
P_Blue (P3)	0.500	0.500	0.500
Multinomial	126	210	210

Parameter	Case 1	Case 2	Case 3
P1^X1	0.009277	0.023438	0.052734
P2^X2	0.125	0.015625	0.125
P3^X3	0.062500	0.062500	0.015625
Probability	0.073242	0.022949	0.010986

Note: The probabilities in the three cases illustrate how varying the count of each outcome type affects the multinomial probability while holding the outcome probabilities constant.

3. Plotting and Fitting of Poisson Distribution and Graphical Representation of Probabilities

The Poisson distribution is a discrete probability distribution that quantifies the probability of a given number of events occurring within a fixed time interval or spatial region. The distribution assumes that events occur continuously and independently at a constant average rate. It is governed by a single parameter, λ (lambda), which represents the average number of event occurrences within the specified interval.

Poisson Distribution Formula

$$P(k) = e^{(-\lambda)} * \lambda^k / k!$$

Mean: $\mu = E(X) = \lambda$

Variance: $\sigma^2 = V(X) = \lambda$

Standard Deviation: $\sigma = \sqrt{\lambda}$

Where:

- $P(X = k)$ is the probability of observing exactly k events
- e is Euler's number (the base of the natural logarithm), approximately 2.71828
- λ is the mean number of events occurring during a specific interval; $\lambda = np$
- k is the number of events (successes) of interest

How to Implement in Excel

Excel Formula: =POISSON.DIST(k, λ, FALSE)

The table below shows the Poisson distribution for λ = 8 (mean number of successes), evaluated up to k = 12:

k (Successes)	Poisson Distribution
0	0.000335463
1	0.002683702
2	0.010734807
3	0.028626019
4	0.057252038
5	0.091603261

k (Successes)	Poisson Distribution
6	0.122137681
7	0.139585921
8	0.139585921
9	0.123854597
10	0.099083678
11	0.072060493
12	0.048040329

When $\lambda = 8$, the distribution peaks around $k = 7$ and $k = 8$, after which the probabilities decrease steadily.

4. Plotting and Fitting of Geometric Distribution and Graphical Representation of Probabilities

In a Bernoulli trial, a geometric distribution models the number of consecutive failures that are observed before the first success is recorded. A Bernoulli trial is defined as a random experiment that can produce exactly one of two outcomes: success or failure. The distinguishing characteristic of a geometric distribution is that the process of repeated Bernoulli trials continues until a success is obtained, at which point it is terminated.

The geometric distribution is a discrete probability distribution that captures the likelihood of needing exactly k failures before the first success occurs. The number of attempts may theoretically extend without bound until the initial success is achieved. This distribution rests on three fundamental assumptions:

- *The individual trials being conducted are mutually independent.*
- *Each trial must result in exactly one of two possible outcomes: either a success or a failure.*
- *The probability of success, denoted by p, remains fixed and identical across every trial.*

Geometric Distribution Formula

$$P(X = k) = (1 - p)^k * p$$

Mean: $\mu = E(X) = (1 - p) / p$

Variance: $\sigma^2 = V(X) = (1 - p) / p^2$

Where:

- k: the count of failures that occur prior to the first success
- p: the probability of achieving success on each individual trial

The probability of a successful outcome is represented by p, whereas the probability of failure on any given trial is denoted by q, where $q = 1 - p$. The notation $X \sim G(p)$ signifies that the random variable X follows a geometric probability distribution.

How to Implement in Excel

Excel Formula: = (1 - p)^k * p

The following table presents the geometric distribution for p = 0.3 (probability of success on each trial):

Failures before 1st Success (k)	Geometric Probability
0	0.3000
1	0.2100
2	0.1470
3	0.1029
4	0.0720
5	0.0504
6	0.0353
7	0.0247
8	0.0173
9	0.0121
10	0.0085
11	0.0059

With p = 0.3, the highest probability occurs at k = 0 (immediate first-try success). As k increases, the probability decreases steadily, reflecting the diminishing chance of extended failure streaks.

5. Plotting and Fitting of Uniform Distribution and Graphical Representation of Probabilities

A uniform distribution is a type of probability distribution in which every outcome within a specified range is equally likely to occur. It is also commonly referred to as a rectangular distribution (in the context of the continuous case). The distribution is fully characterized by two parameters: a (the minimum value) and b (the maximum value), and is expressed as U(a, b).

In a uniform distribution, every possible value within the defined range carries an identical probability of being observed. This property ensures that no value within the interval [a, b] is inherently more likely than any other.

Uniform Distribution Formula

The probability density function (PDF) of a continuous uniform distribution defines the probability that a random variable falls within a particular sub-interval of [a, b]:

$$f(x) = 1 / (b - a) \quad \text{for} \quad a \leq x \leq b$$

$$\text{Mean:} \quad \mu = (a + b) / 2$$

$$\text{Variance:} \quad \sigma^2 = (b - a)^2 / 12$$

How to Implement in Excel

Excel Formula for Probability: P = (x2 - x1) / (b - a)

To compute the probability that a uniform random variable falls between x1 and x2, the following four inputs are required:

1. a: the minimum value of the distribution
2. b: the maximum value of the distribution
3. x1: the lower bound of the interval of interest
4. x2: the upper bound of the interval of interest

The following table presents three cases using a = 10 and b = 30:

Parameter	Case 1	Case 2	Case 3
a (Min)	10	10	10
b (Max)	30	30	30

Parameter	Case 1	Case 2	Case 3
x1	12	16	20
x2	15	20	25
Uniform Distribution P	0.15	0.20	0.25

As shown in the table, the probability is directly proportional to the width of the sub-interval (x2 - x1) relative to the total range (b - a).

6. Plotting and Fitting of Exponential Distribution and Graphical Representation of Probabilities

The exponential distribution is a continuous probability distribution defined over the set of all non-negative real numbers. It is particularly suited for modeling the time elapsed before a given event occurs. When we ask questions such as 'How long must we wait until an event happens?', the answer can be modelled probabilistically using the Exponential Distribution. Since the waiting time is unknown in advance, it is treated as a continuous Random Variable. If the probability of the event occurring during any time interval is proportional to the length of that interval, the Random Variable follows an exponential distribution.

This distribution is commonly applied to the following types of real-world problems:

- How long does a shopkeeper need to wait before a customer enters the store.
- How long will a battery continue to function before it fails.
- How long will a computer system remain operational before it encounters a breakdown.

Exponential Distribution Formula

$$f(x) = \lambda * e^{(-\lambda x)} \quad \text{for } x \geq 0$$

$$f(x) = 0 \quad \text{for } x < 0$$

$$E(X) = 1/\lambda$$

$$Var(X) = 1/\lambda^2$$

Where λ is the rate parameter, which controls the shape and spread of the density function. The constant e is approximately 2.718.

How to Implement in Excel

Excel Formula: =EXPON.DIST(X, lambda, FALSE)

The table below presents the exponential PDF values for μ = 500 and λ = 0.002 across three different x values:

Parameter	Case 1	Case 2	Case 3
mu (μ)	500	500	500
lambda (λ)	0.002	0.002	0.002
x	600	650	700

Parameter	Case 1	Case 2	Case 3
Exponential Distribution	0.001212	0.001097	0.000993

As x increases beyond the mean μ = 500, the exponential PDF decreases progressively, consistent with the decaying nature of the exponential density function.

7. Plotting and Fitting of Normal Distribution and Graphical Representation of Probabilities

The Normal Distribution, also known as the Gaussian distribution, is defined as the probability density function of any continuous random variable within a given system. It describes the distribution of a random variable X whose behavior conforms to a characteristic bell-shaped curve. The Normal distribution is entirely defined by two parameters: the mean (μ) and the standard deviation (σ).

Normal Distribution Formula

$$f(x) = [1 / (\sigma\sqrt{(2\pi)})] * e^{[-(1/2) * ((x - \mu)/\sigma)^2]}$$

Where:

- x is the Random Variable
- μ is the Mean of the distribution
- σ is the Standard Deviation

Properties of Normal Distribution

- For normally distributed data, the mean, median, and mode are all equal (Mean = Median = Mode).
- The total area enclosed beneath the normal distribution curve is always equal to 1.
- The normal distribution curve is symmetric about its central axis, which passes through the mean.
- Exactly half the values in a normally distributed dataset lie to the left of the mean, and the remaining half lie to the right.
- The Normal distribution is completely characterized by just two values: the mean and the standard deviation.
- The normal distribution curve is unimodal, meaning it has exactly one peak at the mean.

How to Implement in Excel

5. Input your complete dataset into an Excel spreadsheet.
6. Compute the mean of the dataset: =AVERAGE(cell range)
7. Compute the standard deviation of the dataset: =STDEV(cell range)
8. Select the value(s) for which you wish to compute the normal probability.
9. Apply the NORM.DIST function: =NORM.DIST(x, mean, standard_deviation, FALSE)

The following table illustrates the normal distribution applied to employee incentive data. The distribution is parameterized with a mean of 12,500 and a standard deviation of 4,200:

Mean	12500
Standard Deviation	4200.00

Employee	Incentive (₹)	Normal Distribution
EMP1	1000	1.31E-05
EMP2	2000	1.65E-05
EMP3	3000	2.02E-05
EMP4	4000	2.42E-05
EMP5	5000	2.83E-05
EMP6	6000	3.24E-05
EMP7	7000	3.63E-05
EMP8	8000	3.98E-05
EMP9	9000	4.27E-05
EMP10	10000	4.49E-05
EMP11	11000	4.61E-05
EMP12	12000	4.63E-05
EMP13	13000	4.55E-05
EMP14	14000	4.37E-05
EMP15	15000	4.10E-05
EMP16	16000	3.76E-05
EMP17	17000	3.36E-05
EMP18	18000	2.94E-05
EMP19	19000	2.51E-05
EMP20	20000	2.10E-05

The values peak near the mean (EMP12-EMP13, incentive ~12,000-13,000) and taper symmetrically in both directions, consistent with the bell-curve shape.

8. Calculation of Cumulative Distribution Functions for Exponential and Normal Distributions

The cumulative distribution function (CDF) gives the probability that a random variable takes a value less than or equal to a specified threshold. The Excel functions for CDF computation use the argument TRUE (cumulative = TRUE) instead of FALSE.

Normal CDF in Excel: =NORM.DIST(x, mean, standard_deviation, TRUE)

Exponential CDF in Excel: =EXPON.DIST(x, lambda, TRUE)

Normal Distribution CDF (mean = 5500, SD = 3027.65):

Employee	Incentive	CDF Normal
EMP1	1000	0.001429
EMP2	2000	0.003985
EMP3	3000	0.009815
EMP4	4000	0.022007
EMP5	5000	0.046070
EMP6	6000	0.089686
EMP7	7000	0.162543
EMP8	8000	0.274253
EMP9	9000	0.415808
EMP10	10000	0.566116

Exponential Distribution CDF (λ = 0.002):

Days	Exp CDF (λ=0.002)
300	0.451188
320	0.472891
350	0.503415
400	0.550671
420	0.568187
450	0.593430

Days	Exp CDF (λ=0.002)
500	0.632121
520	0.647952
550	0.667129
600	0.698806

The CDF values for both distributions are monotonically non-decreasing, approaching 1.0 as x tends to infinity.

9. Given Data from Two Distributions, Find the Distance Between the Distributions

Euclidean Distance

The Euclidean distance measures the straight-line separation between two points (or vectors) in a real-valued space. It is computed as the square root of the sum of the squared element-wise differences between two vectors.

$$\text{Euclidean Distance} = |X - Y| = \sqrt{[\ \Sigma(i=1 \text{ to } n)\ (x_i - y_i)^2\ ]}$$

Where:

- X: Array or vector X
- Y: Array or vector Y
- xi: Values along the horizontal axis in the coordinate plane
- yi: Values along the vertical axis in the coordinate plane
- n: Number of observations

How to Implement in Excel

Excel Formula: =SQRT(SUMXMY2(array_X, array_Y))

The following table computes the Euclidean distance between the Binomial distribution (n = 10, p = 0.5) and the Poisson distribution (k = 10, λ = 0.5) across 10 trial values:

Trials	Binomial Dist.	Poisson Dist.	Euclidean Distance
1	0.00977	0.00674	0.00303
2	0.04395	0.03369	0.01026
3	0.11719	0.08422	0.03297
4	0.20508	0.14085	0.06423
5	0.24609	0.17535	0.07074
6	0.20508	0.14621	0.05887
7	0.11719	0.07310	0.04409
8	0.04395	0.02280	0.02115
9	0.00977	0.00380	0.00597
10	0.00098	0.00034	0.00064

As the number of trials increases, both distributions converge, resulting in progressively smaller Euclidean distances.

10. Application Problems Based on the Binomial Distribution

Problem

Antibiotics occasionally produce nausea as an unwanted side effect. A major pharmaceutical company has developed a new antibiotic under the brand name Phe-Mycin. The manufacturer asserts that, at most, 10 percent of all patients treated with Phe-Mycin would experience nausea as a side effect.

Suppose that a random sample of n = 4 patients is selected and treated with Phe-Mycin. Each patient will either experience nausea (classified as a success) or will not (a failure). We assume that p = 0.10 is the true probability of a patient experiencing nausea, as stated by the manufacturer. It is further assumed that patients' reactions to the drug are mutually independent.

Let x denote the count of patients among the four who experience nausea as a side effect. The variable x is therefore a binomial random variable taking possible values of 0, 1, 2, 3, or 4. Suppose that in a random sample of n = 4 patients, three of them experience nausea. Since 3/4 = 0.75, which is substantially greater than 0.10, this provides evidence that the true probability p may exceed the manufacturer's claimed maximum.

Distribution Parameters: n = 4, p = 0.10

k (Successes)	Binomial Distribution
0	0.6561
1	0.2916
2	0.0486
3	0.0036
4	0.0001

From the table, P(X = 3) = 0.0036, which is a very low probability under the assumption that p = 0.10. This supports the hypothesis that the true nausea rate is higher than 10%.

11. Application Problems Based on the Poisson Distribution

Problem

In a busy café, customers arrive at an average rate of 2 per minute. Determine the probability that exactly 5 customers arrive within a 1-minute time window, using the Poisson distribution formula.

Distribution Parameters: $\lambda = 2$, $x = 5$

x (Customers)	Poisson Distribution
1	0.270671
2	0.270671
3	0.180447
4	0.090224
5 (Answer)	0.036089
6	0.012030
7	0.003437
8	0.000859

Solution: $P(X = 5) = e^{(-2)} \times 2^5 / 5! \approx 0.036089$

The probability of exactly 5 customers arriving in 1 minute, given an average arrival rate of 2 per minute, is approximately 3.61%.

12. Application Problems Based on the Normal Distribution

Problem

According to the American Association for Justice, Stella Liebeck of Albuquerque, New Mexico, sustained severe burns covering 6 percent of her body after receiving McDonald's coffee in February 1992. She was subsequently awarded $160,000 in compensatory damages and $480,000 in punitive damages. A post-verdict investigation revealed that the coffee temperature at the Albuquerque McDonald's location had decreased from approximately 185°F prior to the trial to approximately 158°F after the verdict. This case focuses on the temperatures of coffee served at a fast-food establishment.

To reduce litigation risk and potentially improve the beverage's palatability, the restaurant decided to monitor the temperature of the coffee it serves. To accomplish this, the restaurant staff measured the coffee temperature (in degrees Fahrenheit) at a randomly selected moment during each of the 24 half-hour periods between 8:00 a.m. and 7:30 p.m. on a given day. This procedure was subsequently repeated on a second day, yielding a total of 48 coffee temperature observations.

Mean	155.947
Standard Deviation	5.72008

Temp (°F)	Normal Distribution
146	0.015375
148	0.026567
149	0.033357
150	0.040622
151	0.047981
152	0.054966
153	0.061074
154	0.065817
155	0.068794
156	0.069741
157	0.068573
158	0.065395
159	0.060487
160	0.054264

Temp (°F)	Normal Distribution
161	0.047215
162	0.039845
163	0.032614
164	0.025891
165	0.019936
166	0.014888
167	0.010784
168	0.007576
169	0.005162
171	0.002187
173	0.000820

The normal distribution curve for these coffee temperatures is centered at approximately 155.95°F with a standard deviation of 5.72°F, indicating moderate variability in serving temperatures.

13. Covariance and Correlation — Concept and Computation

Covariance and correlation are two closely related statistical measures used to quantify the relationship between two random variables. While both describe the direction of a linear relationship, they differ in interpretation and scale.

Covariance Formula

$$\mathrm{Cov}(X, Y) = \Sigma[(Xi - \bar{X})(Yi - \bar{Y})] / n$$

Where $\bar{X}$ and $\bar{Y}$ represent the sample means of the two datasets, and n is the number of paired observations.

Key conceptual distinctions between Covariance and Correlation:

10. Both datasets must contain the same number of observations.
11. Neither dataset should be empty, and the standard deviation must not be zero.
12. Covariance quantifies the extent to which two random variables tend to move in tandem.
13. Correlation expresses how strongly two random variables are linearly associated with each other.
14. Covariance is essentially an unscaled measure of correlation.
15. Correlation is the standardized (scaled) form of covariance.
16. Covariance can range from $-\infty$ to $+\infty$, while correlation is bounded between −1 and +1.
17. Covariance identifies only the direction of the linear relationship between variables.
18. Correlation captures both the strength and direction of the linear relationship between two variables.
19. Covariance is sensitive to changes in scale, whereas correlation is scale-invariant.

Pearson Correlation Coefficient Formula

$$r = \Sigma[(xi - \bar{x})(yi - \bar{y})] / \sqrt{[\Sigma(xi - \bar{x})^2 \times \Sigma(yi - \bar{y})^2]}$$

Where:

- r = Pearson correlation coefficient
- xi = individual values of the x-variable in the sample
- $\bar{x}$ = mean of the x-variable values
- yi = individual values of the y-variable in the sample
- $\bar{y}$ = mean of the y-variable values

Types of Correlation

- Positive Correlation: As one variable increases, the other also increases.
- Negative Correlation: As one variable increases, the other decreases.
- No Correlation: The two variables exhibit no discernible linear relationship.

How to Implement in Excel

Pearson Correlation: =PEARSON(array1, array2) or =CORREL(array1, array2)

Population Covariance: =COVARIANCE.P(array1, array2)

Scatter Plots

A scatter plot is a two-dimensional graphical tool that displays the relationship between two numerical variables. Each observation is plotted as a point, with the independent variable on the x-axis and the dependent variable on the y-axis. Scatter plots are also called scatter graphs or scatter diagrams.

Scatter plots are especially useful in the following situations:

- When analyzing large datasets with many paired observations.
- When each data point is represented as a pair of numerical values.
- When both variables are measured on a continuous numerical scale.

Example dataset — Study Hours vs. Marks:

Study Hours	10	13	14	8	5	4	8
Marks	99	87	50	60	70	33	46

The covariance between Study Hours and Marks in this dataset is approximately 34.93, suggesting a positive relationship.

To insert a scatter chart in Excel: navigate to the Insert tab → Charts group → Scatter Chart.

14. Calculation of Karl Pearson's Correlation Coefficients

Karl Pearson's correlation coefficient is a standard measure of the linear association between two variables. It takes values in the range [−1, +1], where −1 indicates a perfect negative linear relationship, 0 indicates no linear association, and +1 indicates a perfect positive linear relationship.

The following dataset presents the values of x alongside three corresponding y-variables (Y1, Y2, Y3):

x	Y1	Y2	Y3
2	80	65	10
5	95	69	30
6	76	60	15
8	58	95	25
10	67	80	10

Correlation	Value
XY1	-0.62732
XY2	0.633225
XY3	0.018149

Interpretation: XY1 shows a moderate negative correlation (−0.627), XY2 shows a moderate positive correlation (+0.633), and XY3 shows virtually no linear correlation (+0.018).

15. Finding the Correlation Coefficient for a Bivariate Frequency Distribution

A bivariate frequency distribution tabulates the joint frequency of two variables simultaneously. It enables the computation of the correlation coefficient directly from the frequency table, providing insight into the linear relationship between the two grouped variables.

The following table presents a bivariate frequency distribution of student Marks (rows) against Age in Years (columns):

Marks \ Age	16-18	18-20	20-22	22-24	Total
10-20	2	1	1	0	4
20-30	3	2	3	1	9
30-40	3	3	5	6	17
40-50	2	2	3	4	11
50-60	0	1	2	2	5
60-70	0	1	2	1	4

Correlation coefficient along the x-axis (Age): −0.18773

Correlation coefficient along the y-axis (Marks): 0.774597

The positive and relatively strong y-axis correlation (0.77) indicates that marks tend to increase with age in years, while the x-axis shows a weak negative correlation.

16. Generating Random Numbers from Discrete Distributions (Bernoulli, Binomial, Poisson)

Random number generation from discrete probability distributions is a core computational task in simulation and statistical modeling. Excel provides built-in functions for generating random values conforming to discrete distributions such as the Binomial.

How to Implement in Excel

For Binomial Distribution: =BINOM.INV(1, P, RAND())

This formula generates a random number equal to 1 with probability P, and 0 otherwise. To generate a random binomial count from n trials, use: =BINOM.INV(n, P, RAND())

Example — Random numbers from a Binomial distribution with n = 10 and p = 0.4:

Random Numbers from Binomial Distribution (n=10, p=0.4)
8
3
5
4
6

Each execution of the formula produces a different random binomial value. These values approximate the distribution B(10, 0.4).

17. Generating Random Numbers from Continuous Distributions (Uniform, Normal)

Continuous distributions such as the Uniform and Normal distributions require different approaches for random number generation. Excel supports this through the NORMINV function combined with the RAND() function.

How to Implement in Excel

For Normal Distribution: =NORMINV(RAND(), mean, standard_deviation)

RAND() generates a uniform random probability between 0 and 1. NORMINV then maps this probability to the corresponding value from the specified Normal distribution.

For Uniform Distribution: =a + RAND() * (b - a)

This generates a uniformly distributed random number between a and b.

Example — Random numbers from a Normal distribution with mean = 20 and SD = 5:

Random Numbers from Normal Distribution (mean=20, SD=5)
17.83214
21.59647
19.34289
23.11874
18.76503

These randomly generated values approximate a Normal distribution $N(20, 5^2)$ and will differ with each worksheet recalculation.

18. Finding the Entropy from a Given Dataset

The entropy of a random variable represents the average degree of information, surprise, or uncertainty embedded in the variable's possible outcomes. For a discrete random variable X that assumes values in the alphabet x and is distributed according to the probability mass function P(x) ∈ [0, 1], the entropy H(X) is defined as:

$$H(X) = -\Sigma \; p(x) * \log_2[p(x)] \quad \text{for } x \in X$$

Important notes on the choice of logarithm base:

- Base 2: yields entropy in units of bits.
- Base e (natural logarithm): yields entropy in natural units (nats).
- Base 10: yields entropy in hartleys (also known as bans or dits).

An equivalent interpretation of entropy is that it equals the expected value of the self-information of a random variable.

The following table computes entropy for coin flip outcomes:

Outcome	P(x)	log2 P(x)	H(x)
HH	1	0	0
HT	0.5	-1	0.5
TH	0.5	-1	0.5
TT	0	0	0

Entropy reaches its maximum value of 1 bit when the coin is fair (P(heads) = P(tails) = 0.5), and falls to 0 when the outcome is certain (probability of 0 or 1).